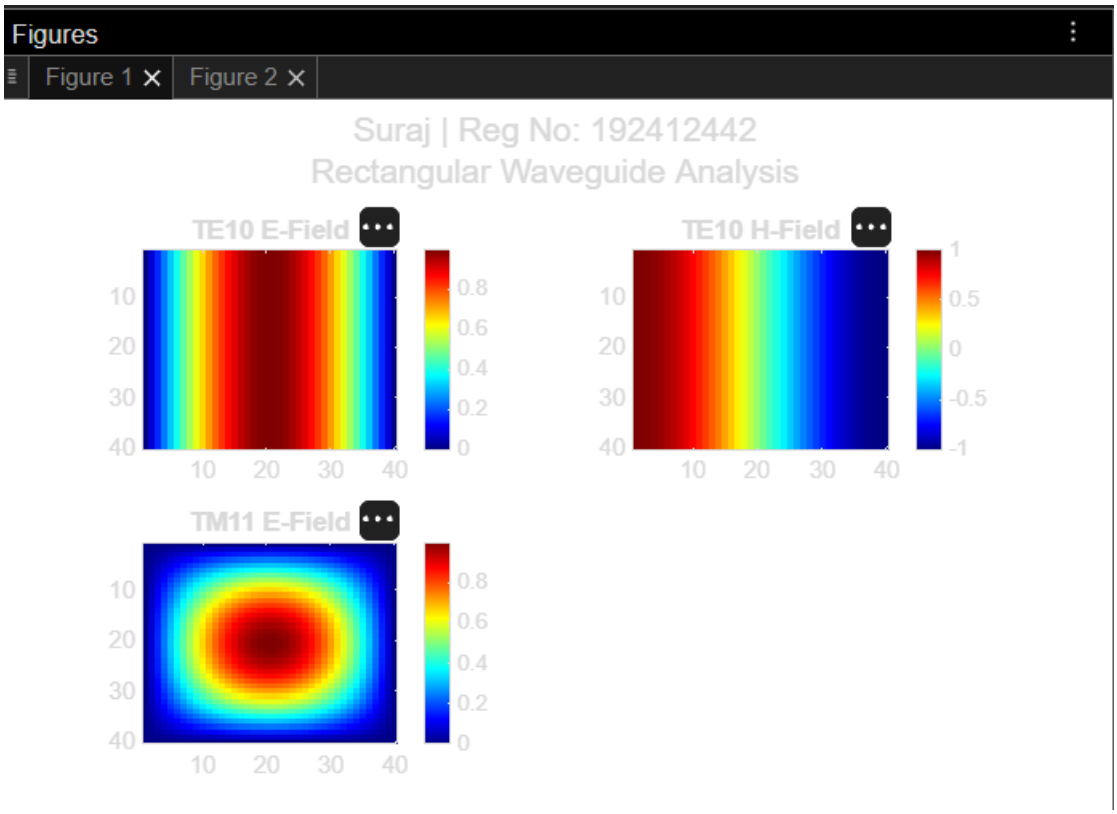
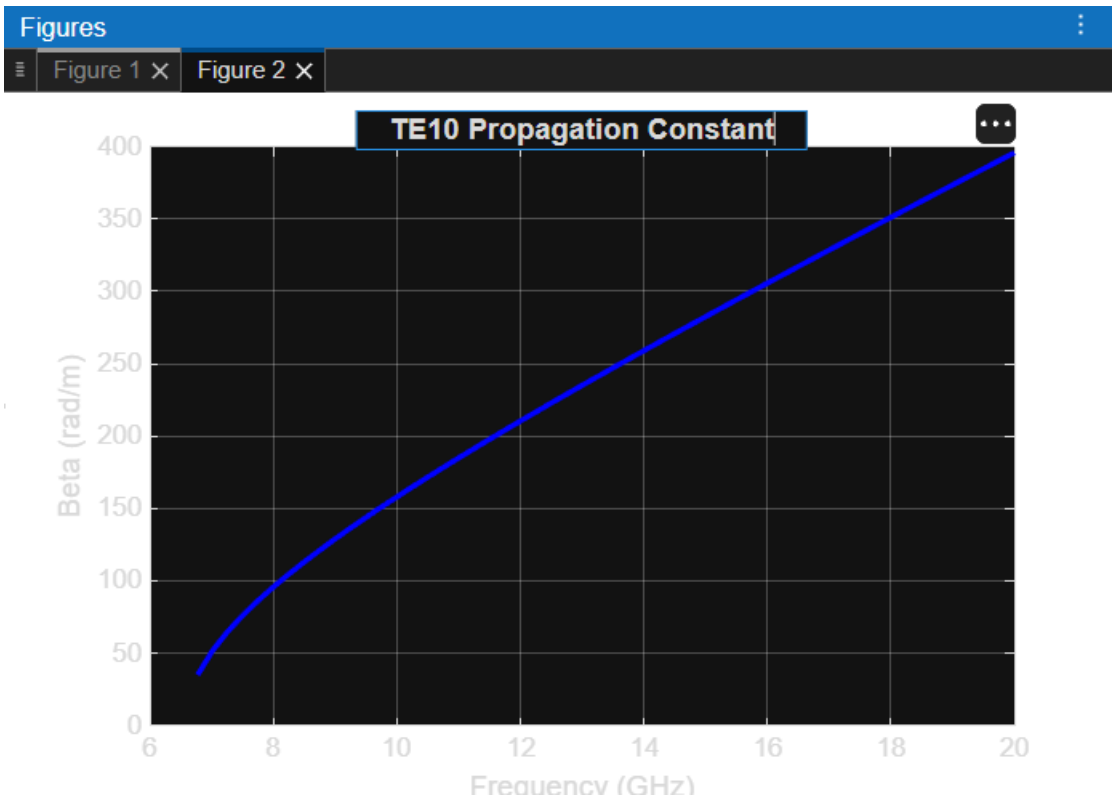
EXPERIMENT 1 (SURAJ NARASIMHAN POTTY_192412442)

CODE:

```
clc; clear; close all;
c=3e8; f=10e9;
a=22.86e-3; b=10.16e-3;
fc=c/(2*a);
ft=c/2*sqrt((1/a)^2+(1/b)^2);
k=2*pi*f/c;
beta=sqrt(k^2-(2*pi*fc/c)^2);
al=sqrt((2*pi*ft/c)^2-k^2);
fprintf('Suraj | Reg: 192412442\n');
fprintf('TE10 fc=%.3f GHz\n',fc/1e9);
fprintf('Beta=%.2f rad/m\n',beta);
x=linspace(0,a,40); y=linspace(0,b,40);
[X,Y]=meshgrid(x,y);
figure('Color','w');
sgtitle({'Suraj | Reg No: 192412442', ...
   'Rectangular Waveguide Analysis'});
subplot(2,2,1); imagesc(sin(pi*X/a)); colorbar; title('TE10 E-Field');
subplot(2,2,2); imagesc(cos(pi*X/a)); colorbar; title('TE10 H-Field');
subplot(2,2,3); imagesc(sin(pi*X/a).*sin(pi*Y/b)); colorbar; title('TM11 E-Field');
subplot(2,2,4); imagesc(sin(pi*X/a)); colorbar; title('TM11 H-Field Pattern');
colormap(jet);
F=linspace(1e9,20e9,80); B=nan(size(F)); q=F>fc;
B(q)=2*pi*F(q)/c.*sqrt(1-(fc./F(q)).^2);
figure('Color','w'); plot(F/1e9,B,'b','LineWidth',2); grid on;
xlabel('Frequency (GHz)'); ylabel('Beta (rad/m)');
title('TE10 Propagation Constant');
```

OUTPUT:


Figures
Figure 1
Figure 2
Suraj | Reg No: 192412442
Rectangular Waveguide Analysis
TE10 E-Field
TE10 H-Field
TM11 E-Field


Figures
Figure 1
Figure 2
TE10 Propagation Constant
Beta (rad/m)
Frequency (GHz)